Evidential Multi-Agent Orchestration for Multimodal Urban Flood Risk Assessment: A Subjective Logic Framework with Provenance-Aware Explanations

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ABSTRACT

Urban flooding is among the most economically destructive categories of natural hazard, with the February–March 2022 south-east Queensland event ranking as one of Australia's costliest flood episodes of recent decades. Effective operational decision-support requires fusing heterogeneous streams — satellite radar, multi-spectral optical imagery, reanalysis precipitation, terrain and land-cover layers, gauge and air-quality stations, and unstructured hydrological bulletins — that arrive at irregular cadences with frequent inter-source disagreement and mutual sensor outages. The downstream decision demands not only a class prediction but also a calibrated, auditable account of which sensors contributed, which were absent, and how strongly the system believes each candidate outcome.

1. Introduction

Urban flooding is among the most economically destructive categories of natural hazard, with the February–March 2022 south-east Queensland event ranking as one of Australia's costliest flood episodes of recent decades. Effective operational decision-support requires fusing heterogeneous streams — satellite radar, multi-spectral optical imagery, reanalysis precipitation, terrain and land-cover layers, gauge and air-quality stations, and unstructured hydrological bulletins — that arrive at irregular cadences with frequent inter-source disagreement and mutual sensor outages. The downstream decision (issuing an inundation warning, prioritizing evacuation, allocating sandbag resources) demands not only a class prediction but also a calibrated, auditable account of which sensors contributed, which were absent, and how strongly the system believes each candidate outcome.

Three architectural families dominate current literature. LLM-orchestrated agentic pipelines delegate sub-tasks to specialist agents and arbitrate conflicts through prompt-chained reasoning [Jiang et al., 2026]. Remote-sensing-grounded multi-agent systems wrap tool-use and retrieval around geospatial foundation models [Talemi et al., 2026, Redaelli et al., 2026]. A third tradition couples dynamic knowledge graphs with environmental inference nodes to mediate between sensors and decision outputs [Hofmeister et al., 2024]. Parallel to this agentic wave sits an evidential-operator literature — multi-source Subjective Logic fusion [van der Heijden et al., 2018], partial-observable projection, credibility-weighted trust models, and evidential deep-learning heads — that is mathematically rigorous but not yet interfaced end-to-end with multi-agent environmental pipelines.

A close reading of these families exposes four structural limitations that block operational deployment. First, LLM-mediated arbitration produces uncalibrated probabilities: LLM-arbiters attain moderate ranking quality but collapse minority-class recall, generating a confident-but-wrong majority-class skew. Second, missing modalities are silently recovered or hallucinated, with no formal monotonicity guarantee on epistemic uncertainty under sensor dropout. Third, per-agent credibility is not propagated — a historically unreliable agent is weighted identically to a historically reliable one. Fourth, no source-provenance trail exists: post-hoc feature attribution explains feature importance but not which sensor or document caused a contribution, so an analyst cannot audit which data source was stale, missing, or uncalibrated. These four limitations motivate a single framework that combines specialist-agent decomposition, formal evidential reasoning, monotonic missing-modality projection, and source-level provenance.

We propose AEGIS[1] — an open-source framework: Agentic Evidential Geographic Intelligence for Sustainability, a specialist multi-agent framework in which each of five evidential agents — Climate, Satellite, Land-Cover, Air-Quality / Sensor, and Document Intelligence — emits a formal Subjective Logic opinion over a shared four-state Dirichlet frame. A deterministic coordinator inspects inter-agent agreement via a Jensen-Shannon signal. When agents agree, it routes through Consensus & Compromise Fusion; when they disagree, it routes through Weighted Belief Fusion weighted by per-agent credibility γ. Missing modalities automatically trigger a partial-observable projection that guarantees the epistemic uncertainty u grows monotonically as modalities drop, rather than subtracting false confidence. The cell-level fused opinion is concatenated with raw multi-modal features and passed to a hybrid LightGBM evidential

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[1] https://github.com/ali-Hamza817/AEGIS

head that retains the formal uncertainty signal as an input feature while preserving a competitively accurate prediction surface. Every emitted and fused opinion is logged to a DuckDB-backed provenance ledger with manifest IDs and source references, powering an interactive Leaflet dashboard that surfaces per-cell uncertainty, per-agent credibility, and per-source contribution shares.

The remainder of this paper is organized as follows. Section 2 reviews related work across agentic environmental AI, subjective-logic foundations, and conflict-resolution techniques. Section 3 develops the mathematical foundations of AEGIS. Section 4 details the architecture. Section 5 describes the Brisbane 2022 study, datasets, baselines, and metrics. Section 6 reports hypotheses H1–H4 with a mechanistic ablation isolating the contribution of the evidential machinery. Section 7 concludes with limitations, broader applicability to other environmental risk domains, and future work.

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